PROJECT PROPOSAL

Detection of Concept Drift in Industrial Data Streams

A study of statistical, ensemble, and windowing based methods for real time industrial IIoT streams

Submitted by

Roll Number	Name
2023BCS0098	G Jeevan Jadhav
2023BCD0018	K Akhil
2023BCS0089	P Joshua Naik
2023BCS0222	N Mohammed Suhail Ahmed Khan

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1. Problem Statement

Industrial systems, including manufacturing lines, Industrial Internet of Things (IIoT) sensor networks, and cyber-physical monitoring platforms, continuously generate high-velocity data streams whose statistical properties change over time. These changes are driven by equipment wear, firmware or protocol updates, new attack patterns, seasonal effects, or process reconfiguration. This phenomenon, known as concept drift, causes machine learning models trained on historical data to silently degrade in accuracy, creating safety and reliability risks in settings such as predictive maintenance and industrial intrusion detection.

This project investigates how to reliably and efficiently detect concept drift in industrial data streams. It compares statistical and error-rate based detectors against ensemble and windowing based approaches, reproduces one representative method as a baseline, and then proposes an enhancement built on top of it.

Research question: Which concept drift detection strategy provides the best trade-off between detection accuracy (F1/AUC), detection delay, and computational overhead when applied to real-time industrial (IIoT) data streams, and can this trade-off be improved through an enhanced hybrid detector?

2. Base Papers

Paper 1 — Benchmarking Change Detector Algorithms from Different Concept Drift Perspectives (Sakurai et al., 2023, Future Internet)

a. Research question: Which of the widely used statistical change detectors, namely DDM, EDDM, HDDM-A, and HDDM-W, performs best across abrupt, gradual, and incremental drift types, and what trade-offs exist between detection accuracy, detection time, and detection delay?

b. Data used: Synthetic binary data streams generated by a custom open-source stream generator that independently controls drift type (abrupt, gradual, incremental), stream size, number of drifts, and drift duration or amplitude, allowing controlled and repeatable experiments across hundreds of streams.

c. Metrics used: F1 score (precision and recall based detection accuracy), detection time (processing time per stream), and detection delay (time between actual drift onset and detector alarm), validated statistically using the Friedman test with Nemenyi post-hoc analysis.

d. Key findings: HDDM-W gave the best overall trade-off, showing the most consistent F1 scores for abrupt drift and the lowest detection delay for both abrupt and gradual drift, though it was not the fastest and struggled with incremental drift. EDDM was statistically the most accurate detector overall and excelled specifically at incremental drift, while DDM and HDDM-A had the fastest detection times but weaker incremental drift detection. No single detector dominated on every axis, confirming the need for scenario-aware detector selection.

e. Model: No predictive or classification model is trained; the paper evaluates four statistical drift detection algorithms (DDM, EDDM, HDDM-A, HDDM-W) directly on binary drift-indicator streams using the River streaming ML library. This is the paper our team will reproduce as a baseline, since its synthetic stream generator and evaluation protocol (F1, detection time, detection delay) are directly reusable for benchmarking on our chosen industrial dataset.

Paper 2 — A Multi-Stage Automated Online Network Data Stream Analytics Framework for IIoT Systems (Yang & Shami, 2022/2023, IEEE Transactions on Industrial Informatics)

a. Research question: How can an end-to-end online analytics pipeline detect concept drift in real IIoT network traffic and automatically re-select features and retrain an ensemble model to sustain high detection accuracy under drift?

b. Data used: Two public IIoT and network intrusion benchmark datasets, IoTID20 and CICIDS2017, used to simulate realistic industrial network traffic with evolving attack behaviour.

c. Metrics used: Classification accuracy, F1 score, per-packet inference time (ms), and throughput (samples per second), benchmarked against other online ensemble methods (SRP, LB, PWPAE).

d. Key findings: The proposed Multi-Stage Automated Network Analytics (MSANA) framework, which pairs ADWIN and EDDM as dual drift detectors with drift-triggered feature re-selection and an online ensemble, achieved 99.32% accuracy and 97.95% F1 on CICIDS2017 with only 3.53 ms inference time per packet, and 98.88% accuracy on IoTID20 with markedly lower latency (2.81 ms) and higher throughput than the closest competing method (PWPAE), demonstrating that drift-triggered adaptation preserves accuracy while cutting inference cost.

e. Model: A four-stage pipeline consisting of dynamic data pre-processing, drift-based dynamic feature selection, base model learning and selection, and a window-based weighted probability averaging ensemble (W-PWPAE), using ADWIN and EDDM as the drift detectors that trigger retraining.

Paper 3 — A Lightweight Concept Drift Detection and Adaptation Framework for IoT Data Streams (Yang & Shami, 2021)

a. Research question: Can a lightweight combination of a gradient-boosting classifier (LightGBM) and a novel adaptive windowing method detect and adapt to concept drift in resource-constrained IoT and industrial environments without sacrificing accuracy?

b. Data used: Two public IoT and network security datasets, IoTID20 and NSL-KDD, with concept drift artificially introduced by manipulating test-set sample distributions to create sudden and gradual drift scenarios. Hardware feasibility was also tested on a Raspberry Pi 3.

c. Metrics used: Accuracy, precision, recall, F1 score, memory usage (MB), and per-sample processing time (ms), benchmarked against SAM-KNN, Adaptive Random Forest (ARF), Streaming Random Patches (SRP), and Hoeffding Anytime Tree (HATT).

d. Key findings: The proposed OASW+LightGBM framework reached 99.92% accuracy on IoTID20 and 98.31% on NSL-KDD, compared with an 84.25% accuracy collapse for a static offline LightGBM model after drift, while using only 31.8 MB of memory and 0.57 ms per sample, substantially less than all competing adaptive methods. This shows that drift-adaptive retraining is both more accurate and cheaper than static or naively retrained models.

e. Model: LightGBM combined with the authors' novel Optimized Adaptive and Sliding Windowing (OASW) drift detection and retraining algorithm, tuned via Particle Swarm Optimization.

3. Dataset / Stream Source

The team will use a combination of two sources:

- The synthetic stream generator from Paper 1, which allows abrupt, gradual, and incremental drift with controllable amplitude and duration, for controlled baseline reproduction.
- A real industrial-style benchmark, IoTID20 (or CICIDS2017), used in Papers 2 and 3, to validate detector behaviour on realistic IIoT traffic exhibiting both naturally occurring and injected drift.

4. Metrics

- Detection accuracy: F1 score, measuring the precision and recall of correctly flagged drift points.
- Detection delay: the time or sample lag between true drift onset and the detector's alarm.
- Detection time / inference latency: per-sample or per-packet processing time (ms), critical for real-time industrial deployment.
- Resource efficiency: memory footprint (MB), relevant for edge and IIoT deployment constraints.
- Downstream classification accuracy: accuracy and F1 of the adaptive model after drift-triggered retraining, to assess whether detection translates into sustained model performance.

5. Chosen Tool Combination

- River (Python streaming ML library): implements DDM, EDDM, HDDM-A, HDDM-W, and ADWIN out of the box, and is used to reproduce Paper 1's baseline detectors.
- LightGBM: a lightweight gradient boosting classifier used for the downstream predictive model, following Paper 3's efficiency-oriented design.
- Scikit-learn: used for the Adaptive Random Forest and evaluation utilities, ensemble baselines, and standard classification metrics (F1, precision, recall).
- Custom synthetic stream generator, reproduced from Paper 1's methodology, to generate controlled abrupt, gradual, and incremental drift scenarios for systematic benchmarking.

6. Expected Enhancement

Building on the baseline formed by Paper 1's DDM, EDDM, and HDDM family of detectors, and the industrial adaptation pipelines described in Papers 2 and 3, the team proposes a hybrid dual-detector drift trigger. This pairs a fast error-rate detector (DDM or HDDM-W, strong on abrupt drift) with a distribution-based detector (EDDM, strong on incremental and gradual drift) running in parallel, similar in spirit to Paper 2's ADWIN plus EDDM combination but tuned for lower false-alarm rates. The enhancement will be evaluated on both the synthetic benchmark, for controlled ground-truth comparison, and the IoTID20 industrial stream, for real-world validity, with the goal of improving detection delay and F1 score over any single baseline detector while keeping per-sample latency within the sub-millisecond range demonstrated in Paper 3.

References

[1] Sakurai, G. Y., Lopes, J. F., Zarpelão, B. B., & Barbon Junior, S. (2023). Benchmarking Change Detector Algorithms from Different Concept Drift Perspectives. *Future Internet, 15(5), 169.* https://doi.org/10.3390/fi15050169

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